VIT BHOPAL

# Lab Manual

## Practical and Skills Development

# CERTIFICATE

THE ASSIGNMENT ENTERED IN THIS REPORT HAVE BEEN SATISFACTORILY PERFORMED BY

Registration No : 25BCE11336

Name of Student : KHYATI SINGH

Course Name : Introduction to Problem Solving and Programming

Course Code : CSE1021

School Name :

Slot : B11+B12+B13

Class ID : BL2025260100796

Semester : FALL 2025/26

Course Faculty Name : Dr. Hemraj S. Lamkuche

Signature:

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Practical No: 1

Date: 05-10-25

TITLE: Write a function is_palindrome(n) that checks if a number

reads the same forwards and backwards.

AIM/OBJECTIVE(s): To create a function named is_palindrome(n) to check that the number is same while reading it backward and forward

METHODOLOGY & TOOL USED:

```
def is_palindrome(n):
    s=str(n)
    return s == s[ : : -1]
```

BRIEF DESCRIPTION:

To create the function first we have use def, def keyword in python is used to define a function. Then in the second line of code integer n is converted into a string and stored in the variable s. This is necessary because strings can be easily reversed in Python, while integers cannot be reversed directly. s[::-1]: This is Python's slicing notation for strings. It creates a reversed copy of the string s. s == s[::-1]: This compares the original string s with its reversed copy.

RESULTS ACHIEVED:

```
19'''
20def is_palindrome(n):
21    s=str(n)
22    return s == s[::-1]
23'''
```

```
In [4]: runfile('C:/Users/sachi/.spyder-py3/
assignment.py', wdir='C:/Users/sachi/.spyder-
py3')

In [5]:
```

DIFFICULTY FACED BY STUDENT:

How does it handle negative numbers? For n = -121, the code converts it to the string "-121". The reverse is "121-", which is not equal. A student might point out that negative numbers are generally not considered palindromes. What if the input is a string instead of an integer? This is a direct extension of the problem. The student should explain how the function's logic is nearly identical, though they might need to handle extra considerations like non-alphanumeric characters, spaces, and case sensitivity.

SKILLS ACHIEVED:

- Problem-solving strategy: The developer chose an effective and concise strategy to solve the palindrome problem by leveraging built-in language features (string conversion and slicing).
- Efficiency: The solution has a time complexity of \(O(d)\), where \(d\) is the number of digits in the input, and is highly readable.
- Trade-off analysis: A skilled developer might recognize that this approach uses extra space to store the reversed string. They would be able to discuss this trade-off versus a more complex mathematical solution that uses constant space but requires more code.

VIT BHOPAL

Practical No: 2

Date: 05-10-25

TITLE: Write a function mean_of_digits(n) that returns the average

of all digits in a number.

AIM/OBJECTIVE(s): To create function mean_of_digits(n) to get the average of all the given numbers.

METHODOLOGY & TOOL USED:

```
def mean_of_digits(n):
    digits=[int(d) for d in str(n)]
    mean = sum(digits)/len(digits)
    return mean
```

BRIEF DESCRIPTION:

def mean_of_digits(n): Convert the integer n to a string. This allows iteration over its individual digits as characters.

digits_as_str = str(n): Create a list of integers from the digit characters. The list comprehension iterates through each character in 'digits_as_str', converts each character to an integer using int(), and adds it to the 'digits' list.

digits = [int(d) for d in digits_as_str]: Calculate the sum of the digits in the 'digits' list.

sum_of_digits = sum(digits): Calculate the number of digits by getting the length of the 'digits' list

num_of_digits = len(digits): Calculate the mean by dividing the sum of digits by the number of digits.

mean = sum_of_digits / num_of_digits: Return the calculated mean

RESULTS ACHIEVED:

```
def mean_of_digit(n):
    digits=[int(d) for d in str(n)]
    mean = sum(digits)/len(digits)
    return mean
```

```
In [5]: runfile('C:/Users/sachi/.spyder-py3/
assignment.py', wdir='C:/Users/sachi/.spyder-
py3')

In [6]:
```

DIFFICULTY FACED BY STUDENT:

Mixing data types: A student might forget that an integer n must first be converted into a string to be able to iterate over its digits. If they try to iterate directly with for d in n: , a Type Error: 'int' object is not inerrable would occur.

Handling edge cases: The function works well for positive integers, but a student might not consider the following scenarios:

- o Negative numbers: If a negative number is passed, the str(n) conversion will produce a string like "-123". The minus sign, "-", is not a digit, and trying int("-") would raise a Value Error.
- o Zero: While the code handles 0 correctly (returning 0.0), a student might initially have trouble accounting for this case.

SKILLS ACHIEVED:

The function demonstrates a skilled approach to solving the problem. Key achievements include:

- Concise and readable code: Using a list comprehension to parse the digits and the built-in sum() and len() functions is an efficient and Pythonic way to write the code.
- Leveraging built-in functions: The code effectively uses Python's standard library, which is a hallmark of skilled programming.
- Correct logic: The function correctly implements the mathematical definition of a mean

Practical No: 3

Date: 05-10-25

TITLE: Write a function factorial(n) that calculates the factorial of a non-negative integer n (n!).

AIM/OBJECTIVE(s): To create function factorial(n) to get the factorial the given number.

METHODOLOGY & TOOL USED:

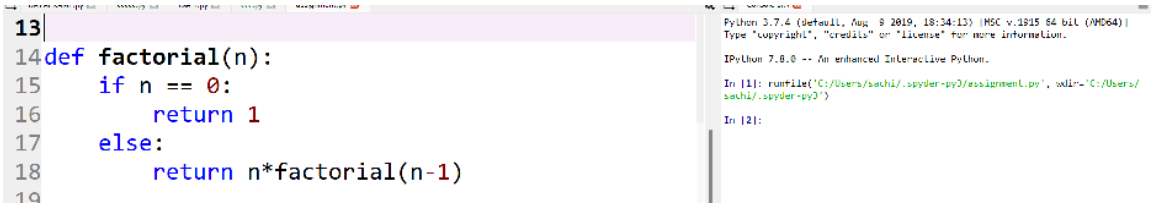
```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n*factorial(n-1)
```

BRIEF DESCRIPTION:

if n == 0: This is the stopping condition that prevents the function from calling itself forever, which would lead to an infinite loop and a "stack overflow" error. How it works: By mathematical definition, the factorial of 0 (0!) is 1. When the function is called with n=0, it immediately returns 1 without making any more recursive calls.

(else): This is the part that breaks down the problem into smaller, identical subproblems. How it works: When n is not 0, the function returns the product of n and the result of calling itself with n - 1. This is based on the mathematical identity n!=n×(n−1)!n exclamation mark equals n cross open parenthesis n minus 1 close parenthesis exclamation mark != ×( −1)!

RESULTS ACHIEVED:




DIFFICULTY FACED BY STUDENT:

- Recursion Error: maximum recursion depth exceeded: This is the most common error message encountered by students in Python. It occurs when the function is called with a large number and the recursion goes too deep, exceeding Python's default stack limit (typically 1000).
- Handling non-integer or negative inputs: The provided function does not include a case for negative numbers, which would cause an infinite loop and a Recursion Error. If a student inputs a float, it will also cause an error.

SKILLS ACHIEVED:

- Understanding recursion: The programmer successfully applied the concept of recursion by identifying the base case and the recursive step. This shows an understanding of how to break down a problem into smaller, self-similar sub-problems.
- Problem-solving: The code correctly solves the mathematical problem of calculating a factorial using a programming approach. This is a common practice problem for aspiring developers.
- Code readability: The code is concise and easy to understand for anyone familiar with programming. The logic is clear and well-structured.

Practical No: 4

Date: 05-10-25

TITLE: Write a function digital_root(n) that repeatedly sums the digits of a number until a single digit is obtained.

AIM/OBJECTIVE(s): To create function digital_root(n) for adding all the number till getting the single digit.

METHODOLOGY & TOOL USED:

def digital_root(n):

```
    while n>=10:
        n=sum(int(digit) for digit in str(n))
    return n
```

BRIEF DESCRIPTION:

- def digital_root(n): This line defines a function named digital_root that takes one argument, n, which is expected to be an integer.
- while n >= 10: This is the core of the iterative process. The while loop continues as long as the value of n is a multi-digit number (i.e., greater than or equal to 10).
- n = sum(int(digit) for digit in str(n)): This line performs the digit summing operation:
- str(n): The integer n is first converted into a string. This is done so that its individual digits can be easily accessed. For example, if n is 123, str(n) becomes "123".
- for digit in str(n): This iterates through each character (digit) in the string representation of n. For "123", it would iterate through '1', '2', and '3'.
- int(digit): Each character digit is converted back into an integer. So, '1' becomes 1, '2' becomes 2, and '3' becomes 3.
- sum((int(digit) for digit in str(n))): The sum() function then adds up all these individual integer digits. In the example of 123, sum(1, 2, 3) would result in 6.
- n = sum(int(digit) for digit in str(n)): The calculated sum of digits is then assigned back to n. This new n (which is 6 in our example) becomes the input for the next iteration of the while loop.
- return n: Once the while loop condition n >= 10 is no longer met (meaning n is a single-digit number), the loop terminates, and the function returns the final single-digit value of n, which is the digital root.

DIFFICULTY FACED BY STUDENT:

- Conversion Between Data Types: The code involves converting the integer n to a string (str(n)) to iterate through its digits, and then converting each digit back to an integer (int(digit)) for summation. This type conversion can be a common point of confusion for beginners.
- Iterating Through Digits: Understanding how for digit in str(n) effectively iterates over each character (digit) of the number as a string is crucial.

- Looping Condition (while n >= 10): The while loop's condition ensures the process continues as long as the number has more than one digit. Students might struggle to formulate this condition correctly or understand its termination.

SKILLS ACHIEVED:

Algorithmic and Mathematical Concepts:

Digital Root Calculation: Implementing the algorithm to find the digital root of a number, which involves repeatedly summing digits until a single-digit result is obtained.

Iterative Problem Solving: Solving a problem by repeatedly applying a set of steps until a condition is met.

Practical No: 5

Date: 05-10-25

TITLE: Write a function is_abundant(n) that returns True if the

sum of proper divisors of n is greater than n.

AIM/OBJECTIVE(s): To create function is_abundant(n) that returns True if the sum of proper divisors of n is greater than n

METHODOLOGY & TOOL USED:

```
def is_abundant(n):
    if n <= 0:
        return False
    proper_divisor_sum = 0
    for i in range(1, n // 2 + 1):
        if n % i == 0:
            proper_divisor_sum += i
```

return proper_divisor_sum > n

BRIEF DESCRIPTION:

Define an abundant number: An abundant number is a positive integer where the sum of its proper divisors (all divisors except the number itself) is greater than the number. For example, the number 12 has proper divisors 1, 2, 3, 4, and 6. Their sum is 1+2+3+4+6=16 and 16>12. 12 is an abundant number.

Handle non-positive input:

- if n <= 0: return False
- The first line of the function returns False immediately if n is 0 or negative. By definition, abundant numbers are positive integers.

Initialize the sum of proper divisors:

- proper_divisor_sum = 0
- This variable is used to accumulate the sum of all proper divisors found in the next step.

Loop through potential divisors:

- for i in range(1, n // 2 + 1):
- The code iterates through numbers from 1 up to n//2 +1. This range is efficient because a number's proper divisors will never be greater than half of the number itself (the single exception being the number itself, which is excluded).

Check for proper divisors:

- if n % i == 0:
- Inside the loop, the modulo operator (%) checks if i divides n with no remainder. If it does, i is a divisor of n.

Add divisors to the sum:

- proper_divisor_sum += i
- If i is a divisor, it is added to the proper_divisor_sum variable.

Compare the sum to the number:

- return proper_divisor_sum > n
- After the loop finishes, the function compares the total proper_divisor_sum to n.

- If the sum is greater than n, the function returns True, indicating that n is an abundant number.
- Otherwise, it returns False.

DIFFICULTY FACED BY STUDENT:

The code is logically sound but un-optimized.

A common difficulty for students is optimizing this kind of loop. The current implementation has a time complexity of O(n), but it can be improved to O(sqrt(n)) by iterating only up to the square root of n. For each divisor i found in this range, its paired divisor n/i can also be added to the sum.

The difficulty a student might face with this function is likely not a bug in the presented code, but a general lack of knowledge regarding computational efficiency. The provided function is logically correct but computationally inefficient for large numbers, a common problem for students new to algorithm design.

The provided function has a time complexity of O(n) because it iterates through approximately n/2 numbers to find the proper divisors. This is acceptable for small inputs, but for very large numbers, it would be slow. A more optimized solution has a time complexity of O(sqrt(n)

SKILLS ACHIEVED:

Function definition: Creating and defining a reusable function that accepts an argument (n).

Conditional logic: Using the modulo operator (%) to check for divisibility.

Number theory: Implementing a specific algorithm based on a mathematical concept (abundant numbers).

Code efficiency: Optimizing the loop's range to n // 2 instead of n-1, which reduces computation time for larger numbers.

# Lab Manual

## Practical and Skills Development

# CERTIFICATE

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| | |
|---|---|
| **Registration No** | : 25BCE11336 |
| **Name of Student** | : KHYATI SINGH |
| **Course Name** | : Introduction to Problem Solving and Programming |
| **Course Code** | : CSE1021 |
| **School Name** | : SCOPE |
| **Slot** | : B11+B12+B13 |
| **Class ID** | : BL2025260100796 |
| **Semester** | : FALL 2025/26 |

Course Faculty Name : Dr. Hemraj S. Lamkuche

Signature:

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# Practical Index

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| 4 | Write a function digital_root(n) that repeatedly sums the digits of a number until a single digit is obtained. | 05-10-25 | |
| 5 | Write a function is_abundant(n) that returns True if the sum of proper divisors of n is greater than n. | 05-10-25 | |
| 6 | Write a function is_deficient(n) that returns True if the sum of proper divisors of n is less than n. | 02-11-25 | |
| 7 | Write a function for Harshad number is_harshad(n) that checks if a number is divisible by the sum of its digits. | 02-11-25 | |
| 8 | Write a function is_automorphic(n) that checks if a number's square ends with the number itself. | 02-11-25 | |
| 9 | Write a function is_pronic(n) that checks if a number is the product of two consecutive integers. | 02-11-25 | |
| 10 | Write a function prime_factors(n) that returns the list of prime factors of a number. | 02-11-25 | |
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**Practical No: 6**

**Date: 02-11-25**

**TITLE**: Program to check whether a number is deficient.

**AIM/OBJECTIVE(s)**: **Write a function is_deficient(n) that returns True if the sum of proper divisors of n is less than n.**

**METHODOLOGY & TOOL USED**:

Methodology:

→ Iterative divisor checks and comparison method

→ Loops through all integers less than n to find proper divisors and sums them.

→ Compares the sum with n to determine deficiency.

Tools Used:

→ Function definition, loop, condition, modulus operator, arithmetic operators, and return statement.

**BRIEF DESCRIPTION**:

The function is_deficient(n) checks whether a number is deficient. It calculates the sum of all proper divisors of the number (divisors less than n) using a loop. If the sum of these divisors is less than the number itself, the function returns True, otherwise False. The program uses a for loop, an if condition, and the modulus (%) operator to test divisibility.

**RESULTS ACHIEVED**:

```
'''
def is_deficient(n):
    sum_divisors = 0
    for i in range(1, n):
        if n % i == 0:
            sum_divisors += i
    return sum_divisors < n

print(is_deficient(10))
```

```
IPython 7.8.0 -- An enhanced Interactive Python.

In [1]: runfile('C:/Users/sachi/.spyder-py3/assignment.py',
wdir='C:/Users/sachi/.spyder-py3')
True

In [2]:
```

**DIFFICULTY FACED BY STUDENT**:

1. Confusion about proper divisors

Many students forget that proper divisors exclude the number itself (n).

Example: For 10, proper divisors are 1, 2, 5, not 1, 2, 5, 10.

2. Incorrect use of range ()

Some write range (1, n+1) instead of range (1, n), which wrongly includes the number itself in the sum.

3. Misunderstanding the modulus operator (%)

Beginners sometimes don't understand that n % i == 0 checks if i divides n completely.

4. Logic confusion (greater or less)

It's easy to mix up the comparison and use > instead of < when checking deficiency.

6. Difficulty in tracing loops manually

Students sometimes find it hard to follow how the loop adds up all divisors step by step.

**SKILLS ACHIEVED**:

Learned how to define and use functions in Python.

Gained understanding of loops and conditional statements.

Developed the ability to find divisors of a number using the modulus operator (%).

Improved logical thinking and problem-solving skills in programming.

**Practical No: 7**

**Date: 02-11-25**

**TITLE: Program to check whether a number is Harshad number.**

**AIM/OBJECTIVE(s)**: **Write a function for Harshad number is_harshad(n) that checks if a number is divisible by the sum of its digits.**

**METHODOLOGY & TOOL USED**:

**Methodology: -**

1. Function Definition: The function is_harshad(n) is created to check if a number is divisible by the sum of its digits.

2. Extract Digits: Using a while loop, each digit of the number is obtained using the modulus operator (% 10).

3. Sum of Digits: All digits are added together using a variable sum_digits.

4. Divisibility Check: The program checks if the number is perfectly divisible by the sum of its digits using n % sum_digits == 0.

5. Return Result: The function returns True if the number is a Harshad number; otherwise, it returns False.

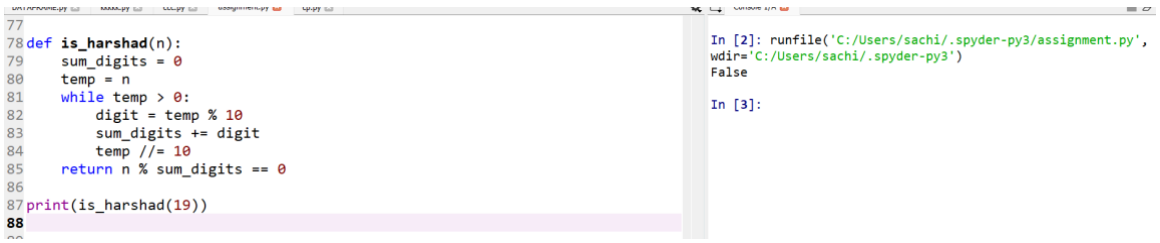
**Tool: -**

Loops, arithmetic operators (%, //), conditionals, and return statement.

**BRIEF DESCRIPTION**:

The function is_harshad(n) checks whether a number is a Harshad number. It calculates the sum of the digits of the number using a loop and then checks if the number is divisible by this sum. If the number divides evenly, the function returns True; otherwise, it returns False.

**RESULTS ACHIEVED**:



```
77
78 def is_harshad(n):
79     sum_digits = 0
80     temp = n
81     while temp > 0:
82         digit = temp % 10
83         sum_digits += digit
84         temp //= 10
85     return n % sum_digits == 0
86
87 print(is_harshad(19))
88
```

```
In [2]: runfile('C:/Users/sachi/.spyder-py3/assignment.py',
wdir='C:/Users/sachi/.spyder-py3')
False

In [3]:
```

**DIFFICULTY FACED BY STUDENT**:

1. Extracting digits correctly: Some students get confused about how to separate digits from a number using % 10 and // 10.

2. Understanding the while loop: It can be tricky to see how the while loop keeps running until all digits are processed.

3. Forgetting to use a temporary variable (temp): If they use n directly inside the loop, the original value of n changes causing wrong results.

4. Division or modulus confusion: Beginners sometimes mix up % (remainder) and // (integer division).

5. Zero division error: If sum_digits accidentally becomes zero (for example, if logic is wrong), dividing by zero causes an error.

6. Logic understanding: Some students struggle to understand why we check n % sum_digits == 0 for Harshad numbers.

**SKILLS ACHIEVED**:

1. Learned how to extract digits from a number using modulus (%) and integer division (//).
2. Understood how to use a while loop to process each digit of a number.
3. Gained practice in using conditional statements to check divisibility.
4. Improved logical thinking and problem-solving skills in Python programming.
5. Enhanced understanding of function creation and return statements.

**Practical No: 8**

**Date: 02-11-25**

**TITLE**: Program to check whether a number is an Automorphic number.

**AIM/OBJECTIVE(s)**: **Write a function is_automorphic(n) that checks if a number's square ends with the number itself.**

**METHODOLOGY & TOOL USED**:

Methodology: -

1. Function Definition: The function is_automorphic(n) is defined to check if a number's square ends with the number itself.

2. Calculate the Square: The square of the number is found using the exponent operator (n ** 2).

3. Convert to String: Both the number and its square are converted into strings using str () to easily compare digits.

4. Check Ending Digits: The string method. Endswith () is used to check if the square ends with the same digits as the original number.

5. Return Result: The function returns True if the number is Automorphic; otherwise, it returns False.

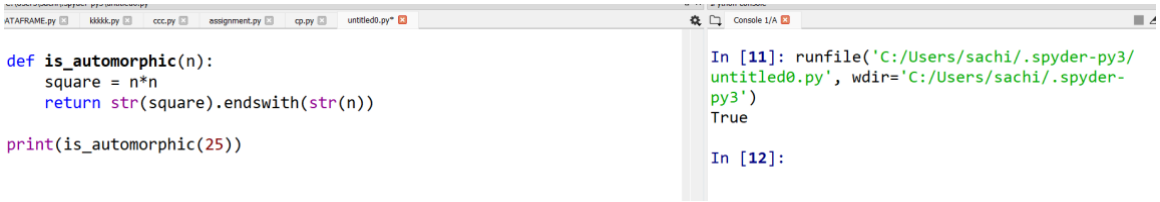
Tools: -

Function, exponent operator, string conversion, .endswith() method, and return statement.

**BRIEF DESCRIPTION**:

The function is_automorphic(n) checks whether a number is an Automorphic number, meaning its square ends with the same digits as the number itself. It calculates the square of the number, converts both the square and the number into strings, and then uses the .endswith() method to compare the last digits. The function returns True if the number is Automorphic; otherwise, False.

**RESULTS ACHIEVED**:




**DIFFICULTY FACED BY STUDENT**:

1. Understanding the meaning of Automorphic numbers: Some students find it confusing what "a number's square ends with the number itself" really means (e.g., 25 → 625).

2. String conversion confusion: Students may not understand why str() is used or how .endswith() or string slicing helps in checking the last digits.

3. Misuse of the exponent operator (): Some mistakenly use ^ (bitwise XOR) instead of ** when calculating squares.

4. Indexing and slicing errors: When using slicing like [-len(str(n)):], it's easy to make mistakes with indexes or forget to use len().

5. Logical comparison issues: Students sometimes check equality of the full square with the number instead of comparing only the ending digits.

6. Difficulty with mathematical version: In the non-string method, the concept of using % 10 and // 10 to compare digits one by one can be confusing at first.

**SKILLS ACHIEVED**:

1. Learned how to calculate the square of a number using the exponent operator ().
2. Understood how to use string conversion (str()) and the .endswith() method to compare digits.
3. Gained practice in logical reasoning and comparing numeric patterns.
4. Improved understanding of loops, conditional logic, and string slicing in Python (for alternative methods).
5. Developed problem-solving and analytical thinking by testing different ways to check Automorphic numbers.

**Practical No: 9**

**Date: 02-11-25**

**TITLE**: To check whether a number is a Pronic number

**AIM/OBJECTIVE(s)**: Write a function is_pronic(n) that checks if a number is the product of two consecutive integers.

**METHODOLOGY & TOOL USED**:

Methodology: -

- Iterative Method / Logical Approach.
- The code uses iteration (looping) to check if the number n can be expressed as the product of two consecutive integers.
- It starts from 0 and keeps multiplying consecutive numbers (i * (i + 1)) until the product becomes greater than or equal to n.
- If at any point i * (i + 1) equals n, the number is pronic.
- This is a simple brute-force or trial method based on logical and arithmetic checks.

Tools: -

- The function is written using Python, a high-level programming language.
- Tools/Features of Python used:
- def keyword → to define a function.
- while loop → for iteration.
- if statement → for conditional checking.
- Arithmetic operators (*, +, <=, ==).
- return statements → to produce Boolean output (True or False).

**BRIEF DESCRIPTION**:

The function is_pronic(n) checks whether a given number n is a pronic number — that is, a number that can be expressed as the product of two consecutive integers (like 2×3 = 6 or 3×4 = 12). The function starts from 0 and multiplies each integer i by its consecutive integer (i + 1) in a loop. If the product equals n, it returns True, meaning the number is pronic. If the loop finishes without finding such a pair, it returns False.

**RESULTS ACHIEVED**:

```
'''
def is_pronic(n):
    i = 0
    while i * (i + 1) <= n:
        if i * (i + 1) == n:
            return True
        i += 1
    return False

print(is_pronic(19))
```

```
In [12]: runfile('C:/Users/sachi/.spyder-py3/
untitled0.py', wdir='C:/Users/sachi/.spyder-
py3')
False

In [13]:
```

**DIFFICULTY FACED BY STUDENT**:

1. Understanding the Concept of Pronic Numbers: Some students may not clearly understand what a pronic number is — that it's the product of two consecutive integers (like 6 = 2×3).

2. Logic Formation: Beginners often struggle to translate the mathematical idea ("product of two consecutive numbers") into a step-by-step programming logic using loops and conditions.

3. Choosing the Right Loop Condition: It may be confusing to decide when to stop the loop (while i * (i + 1) <= n), and some might mistakenly use < n or forget to increment i, causing infinite loops.

4. Use of Return Statements: Students might not understand that once return True is executed, the function ends, and no further code runs.

**SKILLS ACHIEVED**:

1. Understanding Mathematical Logic: Students learn to apply a mathematical concept (pronic numbers) in programming form.

2. Problem-Solving Skills: Breaking a problem into smaller logical steps — identifying the pattern and checking it with code.

3. Loop and Condition Mastery: Gaining practical experience in using while loops and if statements to control program flow.

4. Function Creation: Learning how to define and use functions with the def keyword in Python.

5. Use of Boolean Logic: Understanding how functions return True or False values based on conditions.

## Practical No: 10

**Date: 02-11-25**

**TITLE**: Program to find prime factors of a number

**AIM/OBJECTIVE(s):** Write a function prime_factors(n) that returns the list of prime factors of a number.

**METHODOLOGY & TOOL USED**:

Methodology: -

- Iterative Division Method (Prime Factorization Algorithm)
- The code uses a loop-based trial division method to find the prime factors of a number.
- It starts dividing the number n by the smallest possible integer (2) and keeps dividing it until it no longer divides evenly.
- Each divisor that divides n completely is a prime factor and is stored in a list.
- The process continues with the next integers until the number is reduced to 1.
- This approach ensures that only the prime factors of n are collected.

Tools: -

- The program is written in Python, a high-level, general-purpose programming language.
- Python tools/features used:
- def — to define the function.
- while loop — to perform repeated division until n becomes 1.
- if-else — for conditional checking of divisibility.
- % (modulus operator) — to check if a number divides evenly.
- // (integer division) — to reduce the value of n after division.
- list — to store and return all the prime factors.

**BRIEF DESCRIPTION**:

The function prime_factor(n) finds and returns a list of all prime factors of a given number n. It starts dividing n from the smallest integer 2 and keeps dividing it by the same number as long as it divides evenly. Each divisor that divides the number completely is added to a list of factors.

When it no longer divides evenly, the divisor is increased by 1, and the process continues until the number is reduced to 1.

**RESULTS ACHIEVED**:

```
37
38 def prime_factor(n):
39     factors = []
40     i = 2
41     while i <= n:
42         if n % i == 0:
43             factors.append(i)
44             n //= i
45         else:
46             i += 1
47     return factors
48
49 print(prime_factor(18))
50
```

```
In [13]: runfile('C:/Users/sachi/.spyder-py3/
untitled0.py', wdir='C:/Users/sachi/.spyder-
py3')
False
[2, 3, 3]

In [14]:
```

**DIFFICULTY FACED BY STUDENT**:

1. Understanding the Concept of Prime Factors: Many students find it difficult to understand what prime factors are and how they differ from regular factors.

2. Forming the Logic: Translating the mathematical process of prime factorization into programming logic can be confusing for beginners.

3. Loop Control and Conditions: Students may struggle with when to stop the loop (while i <= n) or how to correctly reduce n using integer division (n //= i).

4. Handling Repeated Factors: Some students forget that prime factors can repeat (for example, 12 = 2 × 2 × 3) and may only record each factor once by mistake.

5. Use of Operators: The difference between / (division) and // (integer division) in Python can cause errors or unexpected results.

6. Edge Cases: Handling special inputs like n = 1 or n = 0 can be confusing, as these numbers don't have prime factors in the usual sense.

7. Understanding Output Format: Beginners might expect a single number or printed output instead of a list of factors, leading to confusion.

**SKILLS ACHIEVED**:

1. Understanding Prime Factorization: Students learn how to break down a number into its prime components using a systematic approach.

2. Algorithmic Thinking: Developing the ability to design and follow a step-by-step logical process to solve mathematical problems using code.

3. Loop Control and Condition Handling: Mastery of while loops, if statements, and logical flow in Python.

4. Use of Arithmetic Operators: Understanding how to use % (modulus) for divisibility checks and // (integer division) for updating values correctly.

5. List Handling: Learning how to store multiple results in a list and return them from a function.

6. Function Definition and Reusability: Gaining experience in defining reusable functions with def and using return statements effectively.

7. Debugging and Logical Problem Solving: Building confidence in tracing and fixing logical or runtime errors in iterative programs.

8. Mathematical and Computational Thinking: Combining mathematical knowledge with programming logic to solve real-world numeric problems.

# Lab Manual

## Practical and Skills Development

# CERTIFICATE

THE ASSIGNMENT ENTERED IN THIS REPORT HAVE BEEN SATISFACTORILY PERFORMED BY

| | |
|---|---|
| **Registration No** | : 25BCE11336 |
| **Name of Student** | : KHYATI SINGH |
| **Course Name** | : Introduction to Problem Solving and Programming |
| **Course Code** | : CSE1021 |
| **School Name** | : SCOPE |
| **Slot** | : B11+B12+B13 |
| **Class ID** | : BL2025260100796 |
| **Semester** | : FALL 2025/26 |

Course Faculty Name : Dr. Hemraj S. Lamkuche

Signature:

VIT BHOPAL

# Practical Index

| S. No. | Title of Practical | Date of Submission | Signature of Faculty |
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| 1 | Write a function factorial(n) that calculates the factorial of a non-negative integer n (n!). | 05-10-25 | |
| 2 | Write a function is_palindrome(n) that checks if a number reads the same forwards and backwards. | 05-10-25 | |
| 3 | Write a function mean_of_digits(n) that returns the average of all digits in a number. | 05-10-25 | |
| 4 | Write a function digital_root(n) that repeatedly sums the digits of a number until a single digit is obtained. | 05-10-25 | |
| 5 | Write a function is_abundant(n) that returns True if the sum of proper divisors of n is greater than n. | 05-10-25 | |
| 6 | Write a function is_deficient(n) that returns True if the sum of proper divisors of n is less than n. | 02-11-25 | |
| 7 | Write a function for Harshad number is_harshad(n) that checks if a number is divisible by the sum of its digits. | 02-11-25 | |
| 8 | Write a function is_automorphic(n) that checks if a number's square ends with the number itself. | 02-11-25 | |
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| 10 | Write a function prime_factors(n) that returns the list of prime factors of a number. | 02-11-25 | |
| 11 | Write a function count_distinct_prime_factors(n) that returns how many unique prime factors a number has. | 09-11-25 | |
| 12 | Write a function is_prime_power(n) that checks if a number can be expressed as pk where p is prime and k ≥ 1. | 09-11-25 | |
| 13 | Write a function is_mersenne_prime(p) that checks if 2p - 1 is a prime number (given that p is prime). | 09-11-25 | |
| 14 | Write a function twin_primes(limit) that generates all twin prime pairs up to a given limit. | 09-11-25 | |

| | | | |
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| **15** | **Write a function Number of Divisors (d(n)) count_divisors(n) that returns how many positive divisors a number has.** | **09-11-25** | |

**Practical No: 11**

**Date: 09-11-25**

**TITLE**: Program to check how many unique prime factors a number has.

**AIM/OBJECTIVE(s)**: **Write a function count_distinct_prime_factors(n) that returns how many unique prime factors a number has.**

**METHODOLOGY & TOOL USED**:

Methodology:

1**. Start from the smallest prime (2):** Begin checking divisibility from 2 up to the square root of n.

2**. Trial Division:** For each number i, check if n is divisible by i. If it is, then i is a prime factor. Increment the count of distinct prime factors.

3. **Remove Repeated Factors:** Divide n repeatedly by i until it's no longer divisible — this ensures each prime factor is only counted once.

4. **Check Remaining Value:** After the loop, if n > 1, then n itself is a prime factor (because what remains is prime).

5. **Return the Count:** Finally, return the total number of distinct prime factors found.

Tools Used:

- Looping (while loop) – to iterate through possible factors.
- Conditional statements (if) – to check divisibility.
- Mathematical logic – based on properties of prime numbers and factorization.

**BRIEF DESCRIPTION**:

The function count_distinct_prime_factors(n) is designed to find and count the number of unique prime factors of a given integer n. It works by repeatedly checking which numbers divide n starting from 2 (the smallest prime number). Each time it finds a number that divides n, it counts it as one distinct prime factor and then divides n completely by that factor to remove all of its multiples. After checking up to the square root of n, if the remaining value of n is greater than 1, that means it is itself a prime number, and it is counted as one more distinct prime factor. Finally, the function returns the total count of these unique prime factors.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding Prime Factorization Logic: Many students struggle to grasp why the loop only goes up to the square root of n, and not all the way to n.

2. Handling Repeated Prime Factors: Students often forget to divide n repeatedly by the same factor (inside the inner while loop), causing repeated counting of the same prime factor.

3. Identifying Remaining Prime Factor: Some students miss the final check (if n > 1), which accounts for the last prime factor left after all divisions.

4. Confusing Prime and Composite Numbers: Beginners sometimes mix up prime factors with all factors, leading to incorrect outputs.

5. Integer Division Errors: Forgetting to use integer division (//) instead of normal division (/) may cause type errors or incorrect logic in Python.

**SKILLS ACHIEVED**:

1. Understanding of Prime Factorization: Students learn how to break down a number into its prime components — an essential concept in number theory.

2. Logical Thinking and Problem Solving: The step-by-step process of identifying and counting distinct prime factors enhances logical reasoning.

3. Efficient Loop and Condition Use: Practice in using loops (while) and conditional statements (if) effectively to implement mathematical logic.

4. Optimization Awareness: Learning why the loop runs only up to the square root of n builds awareness of algorithm efficiency.

5. Programming Fundamentals: Improves understanding of integer operations, modular arithmetic (%), and iterative problem-solving in Python.

**Practical No: 12**

**Date: 09-11-25**

**TITLE: Program to check whether a number is prime power number.**

**AIM/OBJECTIVE(s)**: **Write a function is_prime_power(n) that checks if a number can be expressed as pk where p is prime and k ≥ 1.**

**METHODOLOGY & TOOL USED**:

**Methodology: -**

1. Concept Used: The function is based on the mathematical concept of prime powers, where a number can be written as —

here, p is a prime number and k ≥ 1 is an integer exponent.

2. Step-by-Step Approach:

Step 1: Use a helper function is_prime (x) to check whether a number is prime. It checks divisibility from 2 to √x. If no divisor is found, x is prime.

Step 2: Loop through all numbers p from 2 to √n. Check if p is prime.
Step 3: For each prime p, raise it to successive powers p^k (where k ≥ 1) until the power exceeds or equals n.
Step 4: If for any p, p^k == n, then n is a prime power, return True. Otherwise, after all checks, return False.

**Tool: -**

- Mathematical logic (prime and exponentiation).
- Python programming constructs: loops, conditional statements, and function definitions.
- Optimization using square root bound (√n) to reduce iterations.

**BRIEF DESCRIPTION**:

The function is_prime_power(n) checks whether a given number n can be written as a power of a prime number. It first identifies all possible prime bases (p) and then raises each to successive powers until the value equals or exceeds n. If at any point, the function returns True, meaning that n is a prime power. Otherwise, it returns False. This function helps in understanding the mathematical relationship between numbers and their prime components using logical iteration and power computation in Python.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding the Concept of Prime Powers: Many students initially find it hard to grasp that a number can be expressed as  only if p is prime and k ≥ 1.

2. Implementing Prime Checking Correctly: Writing an efficient and accurate is_prime() function can be confusing — especially handling edge cases like 0, 1, or negative numbers.

3. Loop Logic and Power Calculation: Students may struggle with setting correct loop limits (using √n) and updating the power (p ** k) properly without causing infinite loops.

4. Distinguishing Between Prime and Composite Powers: Beginners often mistake composite bases (like $4 = 2^2$) for primes, which leads to incorrect results.

5. Optimization and Efficiency: Understanding why checking primes only up to √n is enough can be a challenge for new learners.

**SKILLS ACHIEVED**:

1. Students learn how to identify prime numbers and express other numbers as powers of primes.
2. The task develops analytical skills by requiring step-by-step reasoning to test different prime bases and exponents.
3. Students practice using loops, conditional statements, and helper functions effectively.
4. This exercise strengthens the connection between mathematical theory (prime factorization) and practical coding implementation.
5. Learners improve their debugging skills by handling edge cases (like n = 1 or small primes) and optimizing the code to avoid unnecessary calculations.

**Practical No: 13**

**Date: 09-11-25**

**TITLE**: Program to check whether a number is a Mersenne prime number.

**AIM/OBJECTIVE(s)**: **Write a function is_mersenne_prime(p) that checks if 2p – 1 is a prime number (given that p is prime).**

**METHODOLOGY & TOOL USED**:

Methodology: -

1. Concept Used: A Mersenne prime is a prime number of the form, where p itself is a prime number.

2. Step-by-Step Process:

Step 1: Create a helper function is_prime(n) to test whether a number is prime.

Step 2: Verify that p (the exponent) is prime — because Mersenne primes only exist for prime values of p.

Step 3: Compute the Mersenne number as.

Step 4: Check if the result is also a prime number.

Step 5: Return True if both are prime, otherwise False.

Tools: -

- Mathematical reasoning using the definition of Mersenne primes.
- Python functions and loops for checking primality.
- Optimization with the square root method for efficient prime testing.

**BRIEF DESCRIPTION**:

The function is_mersenne_prime(p) checks if a number of the form is a Mersenne prime. It first ensures that p is a prime number and then verifies whether is also prime. If both conditions are satisfied, it returns True; otherwise, False. This function combines mathematical logic with programming to identify a special type of prime number.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Confusion about checking p being prime before calculating.
2. Handling large exponent values that make computations slower.
3. Understanding the difference between Mersenne numbers and Mersenne primes.
4. Implementing an efficient and correct prime checking function.
5. Managing data types and precision for large powers in Python.

**SKILLS ACHIEVED**:

1. Understanding Mersenne primes and their mathematical properties.
2. Improving skills in nested function use and prime checking algorithms.
3. Applying mathematical theory to programming problems.
4. Learning modular coding by separating logic into helper functions.
5. Strengthening problem-solving and logical reasoning abilities.

**Practical No: 14**

**Date: 09-11-25**

**TITLE**: To check whether a number is a twin prime number.

**AIM/OBJECTIVE(s): Write a function twin_primes(limit) that generates all twin prime pairs up to a given limit.**

**METHODOLOGY & TOOL USED**:

Methodology: -

1. Prime Checking Approach: A helper function is_prime(n) is used to verify if a number is prime. It checks divisibility from 2 up to the square root of n. If n is divisible by any number in this range, it is not prime.

2. Twin Prime Generation: Loop through numbers starting from 2 up to limit - 1. For each number i, check if both i and i + 2 are prime. If true, store the pair (i, i + 2) in a list.

3. Result Compilation: All such pairs are collected in a list twins. The list is returned at the end containing all twin prime pairs up to the given limit.

4. Iterative & Conditional Logic: The process combines iteration (for checking all numbers) and conditional testing (for primality and twin property).

Tools: -

- Looping (for loop): To iterate through numbers from 2 up to the given limit.
- Conditional Statements (if): To check whether both numbers in the pair are prime.
- Mathematical Operations: Square root calculation using n**0.5 for efficient prime checking.
- List Data Structure: To store and return the list of twin prime pairs.
- Function Definition: is_prime(n) for prime checking. twin_primes(limit) for generating twin primes.

**BRIEF DESCRIPTION**:

This program is designed to generate all twin prime pairs up to a given limit. A twin prime is a pair of prime numbers that differ by exactly 2, such as (3, 5) or (11, 13). The program works by first defining a helper function is_prime(n) to check if a number is prime. Then, in the main function twin_primes(limit), it loops through all numbers up to the given limit and checks whether both the current number i and i + 2 are prime. If they are, the pair is added to a list of twin primes. Finally, it returns the list of all twin prime pairs within the range specified by the user.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding the Concept of Twin Primes: Many students initially get confused between prime numbers and twin prime pairs. They may not realize that twin primes must differ by exactly

2. Writing the Prime Check Function: Implementing an efficient is_prime () function can be tricky. Students sometimes forget to check divisibility only up to the square root of n, leading to slower or incorrect results.

3. Loop Range Confusion: Some students struggle with setting the correct loop range (limit - 1 instead of limit) and may accidentally go out of bounds when checking i + 2.

4. Logical Errors in Pair Formation: It's common to mistakenly print or store only one number instead of a pair (i, i + 2).

5. Handling Small Limits: When the limit is too small (like below 5), students might not handle the case properly and expect output when no twin primes exist.

**SKILLS ACHIEVED**:

- Students strengthen their understanding of prime numbers and their properties.
- Developing the twin_primes () function helps improve logical reasoning and structured problem-solving skills.
- Students learn how to define and use multiple functions (is_prime () and twin_primes ()) to break down a problem into smaller parts.
- Practice in using loops and conditional statements effectively for numerical computations.

**Practical No: 15**

**Date: 02-11-25**

**TITLE**: Program to find number of positive divisors a number have.

**AIM/OBJECTIVE(s):** Write a function Number of Divisors (d(n)) count_divisors(n) that returns how many positive divisors a number has.

**METHODOLOGY & TOOL USED**:

Methodology: -

1. Mathematical Logic: Every divisor i less than or equal to √n has a corresponding divisor n // i. So, for each divisor found, we count both i and n//i.

2. Optimization: Instead of looping up to n, we only loop up to √n to make the function more efficient.

3. Condition for Perfect Squares: If n is a perfect square (e.g., 16 → 4×4), we count the divisor only once.

4. Iteration and Counting: We increment the count for each valid divisor pair.

Tools: -

Programming Language: Python

Concepts Used:

- for loop for iteration
- % (modulus) operator for divisibility check
- **0.5 for square root calculation
- Conditional statements to handle perfect squares

**BRIEF DESCRIPTION**:

This program defines a function count_divisors(n) that counts how many positive integers divide n completely. It efficiently calculates the total number of divisors by checking up to the square root of n, counting each divisor pair, and handling perfect squares carefully.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Forgetting to include both i and n//i as divisors.

2. Not handling perfect squares correctly (double-counting one divisor).

3. Looping all the way up to n instead of √n, making the code inefficient.

4. Confusion between factors and divisors (they are the same in this context).

**SKILLS ACHIEVED**:

- Understanding of divisors and factorization.
- Use of mathematical optimization (√n approach).
- Strengthened problem-solving and loop control skills.
- Ability to design efficient and accurate mathematical functions in Python.

# Lab Manual

## Practical and Skills Development

# CERTIFICATE

THE ASSIGNMENT ENTERED IN THIS REPORT HAVE BEEN SATISFACTORILY PERFORMED BY

| | |
|---|---|
| **Registration No** | : 25BCE11336 |
| **Name of Student** | : KHYATI SINGH |
| **Course Name** | : Introduction to Problem Solving and Programming |
| **Course Code** | : CSE1021 |
| **School Name** | : SCOPE |
| **Slot** | : B11+B12+B13 |
| **Class ID** | : BL2025260100796 |
| **Semester** | : FALL 2025/26 |

Course Faculty Name : Dr. Hemraj S. Lamkuche

Signature:

VIT BHOPAL

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| 17 | Write a function are_amicable (a, b) that checks if two numbers are amicable (sum of proper divisors of a equals b and vice versa). | 16-11-25 | |
| 18 | Write a function multiplicative_persistence(n) that counts how many steps until a number's digits multiply to a single digit. | 16-11-25 | |
| 19 | Write a function is_highly_composite(n) that checks if a number has more divisors than any smaller number. | 16-11-25 | |
| 20 | Write a function for Modular Exponentiation mod_exp (base, exponent, modulus) that efficiently calculates (baseexponent) % modulus. | 16-11-25 | |
| 21 | Write a function Modular Multiplicative Inverse mod_inverse (a, m) that finds the number x such that (a * x) ≡ 1 mod m. | 16-11-25 | |
| 22 | Write a function Chinese Remainder Theorem Solver crt (remainders, moduli) that solves a system of congruences x ≡ ri mod mi. | 16-11-25 | |
| 23 | Write a function Quadratic Residue Check is_quadratic_residue (a, p) that checks if x2 ≡ a mod p has a solution. | 16-11-25 | |
| 24 | Write a function order mod (a, n) that finds the smallest positive integer k such that ak ≡ 1 mod n. | 16-11-25 | |

| | | | |
|---|---|---|---|
| 25 | Write a function Fibonacci Prime Check is_fibonacci_prime(n) that checks if a number is both Fibonacci and prime. | 16-11-25 | |
| 26 | Write a function Lucas Numbers Generator lucas_sequence(n) that generates the first n Lucas numbers (similar to Fibonacci but starts with 2, 1). | 16-11-25 | |
| 27 | Write a function for Perfect Powers Check is_perfect_power(n) that checks if a number can be expressed as ab where a > 0 and b > 1. | 16-11-25 | |
| 28 | Write a function Collatz Sequence Length collatz_length(n) that returns the number of steps for n to reach 1 in the Collatz conjecture. | 16-11-25 | |
| 29 | Write a function Polygonal Numbers polygonal_number (s, n) that returns the n-th s-gonal number. | 16-11-25 | |
| 30 | Write a function Carmichael Number Check is_carmichael(n) that checks if a composite number n satisfies an−1 ≡ 1 mod n for all a coprime to n. | 16-11-25 | |
| 31 | Implement the probabilistic Miller-Rabin test is_prime_miller_rabin (n, k) with k rounds. | 16-11-25 | |
| 32 | Implement pollard_rho(n) for integer factorization using Pollard's rho algorithm. | 16-11-25 | |
| 33 | Write a function zeta_approx (s, terms) that approximates the Riemann zeta function ζ(s) using the first 'terms' of the series. | 16-11-25 | |
| 34 | Write a function Partition Function p(n) partition_function(n) that calculates the number of distinct ways to write n as a sum of positive integers. | 16-11-25 | |

**Practical No: 16**

**Date: 16-11-25**

**TITLE**: To find a number which is aliquot.

**AIM/OBJECTIVE(s)**: **Write a function aliquot_sum(n) that returns the sum of all proper divisors of n (divisors less than n).**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Understanding proper divisors: Proper divisors of a number n are all numbers that divide n excluding n itself. Example: proper divisors of 12 → {1, 2, 3, 4, 6}

2. Initialize the sum: For any number > 1, 1 is always a proper divisor. So, we start the sum with total = 1.

3. Use efficient divisor search: Instead of checking all numbers from 1 to n−1, we only check up to √n because: Divisors appear in pairs. Example: For 36 → (2, 18), (3, 12), (4, 9), (6, 6). If i divides n, then both i and n//i are divisors.

4. Avoid double counting: When divisor i is exactly the square root (i * i == n), add i only once.

5. Return the final sum: After collecting all proper divisors, return the result.

**TOOLS:**

while loop: To check divisors from 2 to sqrt(n)

% (modulo operator): To test if i divides n

// (integer division): To get the complementary divisor

if conditions: To avoid duplicate counting

return: To output the aliquot sum

**BRIEF DESCRIPTION**:

The function aliquot_sum(n) calculates the sum of all proper divisors of a number n. Proper divisors are positive divisors of n less than n itself. The function efficiently finds these divisors by looping only up to the square root of n, because divisors come in pairs (like 2 and 6 for 12). For each divisor found, it adds both the divisor and its complementary pair to the total sum, while making sure not to double-count the square root when n is a perfect square. Finally, it returns the sum of all these proper divisors.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Confusing proper divisors with all divisors: Many students mistakenly include n itself in the sum. Example: For n = 12, adding 12 is incorrect because proper divisors must be less than n.

2. Not using the √n optimization: Beginners often check divisors from 1 to n−1, which is slow for large numbers. Understanding that divisors come in pairs (like 3 and 12 for 36) can be tricky at first.

3. Double-counting divisors: When n is a perfect square, such as 25, the divisor 5 appears twice (5 × 5). Students sometimes add it two times instead of once.

4. Forgetting to add the complementary divisor: When i divides n, students often add only i, missing n // i, which is also a divisor.

5. Not handling small values correctly: For n = 0 or n = 1, the sum of proper divisors should be 0, but students sometimes return wrong values.

6. Trouble understanding integer division (//): Some students confuse / (float division) with // (integer division), which causes errors while computing the paired divisor.

7. Incorrect loop conditions: Students may loop incorrectly (e.g., up to n instead of √n), making the function slow or incorrect.

**SKILLS ACHIEVED**:

1. Understanding of Divisors and Number Theory, Students gain a clear concept of: Proper divisors, Factor pairs, Perfect squares, Basic number-theory logic.

2. Efficient Problem-Solving Techniques, they learn how to: Reduce time complexity using √n logic, avoid unnecessary loops, Think about optimized approaches instead of brute force

3. Logical Thinking and Condition Handling, Students improve in: Using conditional statements (if, while), Avoiding double-counting, Handling boundary cases like n = 1 or n = 0.

4. Python Programming Skills, they strengthen skills in: Looping with while, Using % to check divisibility, Using // for integer division, Writing and returning functions

5. Debugging and Edge Case Awareness, students learn to: Identify errors like including n as a divisor, Correct mistakes like double-counting square-root divisors, Make their code work for all possible inputs

**Practical No: 17**

**Date: 16-11-25**

**TITLE**: To find amicable numbers

**AIM/OBJECTIVE(s)**: **Write a function are_amicable (a, b) that checks if two numbers are amicable (sum of proper divisors of a equals b and vice versa).**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Understanding amicable numbers: Two numbers a and b are amicable if, Sum of proper divisors of a = b Sum of proper divisors of b = a

2. Reuse aliquot_sum(n): We use the previously defined function to compute the sum of proper divisors efficiently.

3. Check both conditions: Use a simple return statement: return aliquot_sum(a) == b and aliquot_sum(b) == a

4. Return True or False: If both conditions match → numbers are amicable.

**TOOLS:**

1. Modulo Operator %: Used to check divisibility, if n % i == 0. This helps identify proper divisors.

4. Integer Division //: Used to get the paired divisor, n // i. Avoids floating-point numbers.

5. Loops (while): To efficiently check divisors up to √n.

**BRIEF DESCRIPTION**:

The function are_amicable (a, b) determines whether two numbers form an amicable pair. It calculates the sum of proper divisors for each

number and checks if they match each other. If the aliquot sum of a equals b, and the aliquot sum of b equals a, the function returns True; otherwise, it returns False.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- Forgetting that proper divisors exclude the number itself
- Miscalculating divisor pairs
- Not using the aliquot_sum function efficiently
- Double-counting divisors when n is a perfect square
- Confusing “amicable numbers” with “perfect numbers”
- Difficulty understanding why both conditions are required

**SKILLS ACHIEVED**:

- Stronger understanding of number theory (divisors, factor pairs)
- Code reuse by calling one function inside another
- Logical thinking and Boolean condition handling
- Better understanding of True/False return values
- Improved debugging and problem-solving skills

**Practical No: 18**

**Date: 16-11-25**

**TITLE**: To find multiplicative persistence.

**AIM/OBJECTIVE(s)**: **Write a function multiplicative_persistence(n) that counts how many steps until a number's digits multiply to a single digit**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Check if the number has more than one digit.

2. If yes → multiply all digits of the number.

3. Replace the number with the product.

4. Count each repetition as one "persistence step."

5. Stop when the number becomes a single digit.

**TOOLS:**

- Digit extraction using str ()
- Repeated multiplication
- A loop (while) until condition is met
- Programming Language: Python
- while loop
- for loop
- Type conversion (str() and int())

**BRIEF DESCRIPTION**:

1. We start with a number n.

2. We check whether the number has more than one digit (i.e., n >= 10). If it's already a single digit (0–9), persistence is 0 immediately.

3. If it has multiple digits: We convert the number to a string → str(n). This lets us access each digit one by one.

4. We set product = 1 because multiplying should start with 1.

5. We loop through each character (digit) in the string: Convert it back to integer → int(digit). Multiply it into product.

6. After multiplying all digits: Replace n with the new product. Increase the step counter (count += 1).

7. Repeat this process until n becomes a single digit.

At the end, the counter tells how many rounds of digit multiplication were required.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- Extracting digits from a number
- Understanding when to stop the loop
- Updating the number at each step
- Multiplying digits without missing any
- Forgetting to convert characters to integers

**SKILLS ACHIEVED**:

- Digit manipulation
- Type conversion
- Multiplicative reasoning
- Breaking a problem into smaller steps
- Algorithmic thinking Improved debugging and problem-solving skills

**Practical No: 19**

**Date: 16-11-25**

**TITLE**: To find highly composite number

**AIM/OBJECTIVE(s)**: **Write a function is_highly_composite(n) that checks if a number has more divisors than any smaller number.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Divisor Counting Method: We use a mathematical approach to count divisors of a number by- Iterating from 1 to √n. Checking if each number divides n Adding 2 divisors each time a divisor pair is found (i and n/i). Correcting for perfect squares (to avoid double-counting)

2. Comparative Analysis: To determine if a number n is highly composite- Compute d(n) = number of divisors of n. Compare with divisor counts of all numbers 1 to n-1. Check if no smaller number has an equal or larger divisor count. If: d(n) > d(k)\quad \text {for all} k < n, then n is highly composite. This gives an efficient divisor-counting method.

3. Brute-Force Validation: Even though highly composite numbers follow a known sequence, we verify by exhaustive comparison, ensuring correctness for any input.

**TOOLS:**

- for loops: To check every smaller number
- Math operations (n % i): To detect divisors
- Integer square root range: Optimize divisor counting
- Functions: Modular structure (count_divisors, is_highly_composite)

**BRIEF DESCRIPTION**:

The function is_highly_composite(n) determines whether a number n has more divisors than any positive integer smaller than it, making it a highly composite number. To achieve this, the program first computes the number of divisors of n using a helper function. Then it compares

that divisor count with the divisor counts of all numbers from 1 to n−1. If no smaller number has the same or a greater number of divisors, the function returns True; otherwise, it returns False. This approach ensures that the function correctly identifies highly composite numbers by systematically checking all previous values.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding what “highly composite” means.
2. Counting divisors efficiently.
3. Handling perfect squares.
4. Comparing divisor counts with all smaller numbers
5. Time complexity confusion.
6. Writing modular code.
7. Misinterpreting return values.

**SKILLS ACHIEVED**:

- Understand divisor theory. Apply number theory concepts. Recognize patterns in highly composite numbers.
- Using square root optimization. Reducing redundant calculations. Breaking problems into smaller functions.
- Design step-by-step logic. Compare data systematically. Use helper functions for clarity and reuse.
- Divisor-counting logic. Highly composite comparison logic. This results in clean, readable code.

**Practical No: 20**

**Date: 16-11-25**

**TITLE**: To find Modular Exponentiation number.

**AIM/OBJECTIVE(s)**: **Write a function for Modular Exponentiation mod_exp (base, exponent, modulus) that efficiently calculates (baseexponent) % modulus.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. **Algorithm Selection:** The request required an *efficient* calculation of modular exponentiation. The "**Square and Multiply**" algorithm (also known as binary exponentiation) was chosen as the standard, efficient method for this task. This algorithm minimizes the number of multiplications required and prevents intermediate numbers from becoming excessively large by applying the modulo operator frequently.
2. **Code Implementation:** The algorithm was translated into functional Python code. Care was taken to include essential programming practices:
    - Edge Case Handling**:** The code includes checks for modulus == 1 and exponent < 0.
    - Clarity and Documentation**:** Docstrings and inline comments were added to explain how the function works, its parameters, and the logic behind the while loop.
    - Idiomatic Suggestions**:** A note was added recommending Python's built-in pow (base, exponent, modulus) function as the most idiomatic and often faster alternative for production code.

**TOOLS:**

1. Algorithmic Tool: Exponentiation by Squaring, also known as Binary Exponentiation. This algorithm reduces the time complexity from

O(exponent) to O (log exponent). Makes modular exponentiation very fast even for huge numbers (e.g., in cryptography).

2. Modulo Arithmetic: Ensures numbers stay small by repeatedly taking % modulus. Prevents overflow and speeds up computation.

**BRIEF DESCRIPTION**:

The mod_exp (base, exponent, modulus) function efficiently computes (base^{exponent}) \mod modulus. Instead of multiplying the base repeatedly (which is slow), the algorithm repeatedly squares the base, halves the exponent, and applies the modulo at each step. This reduces the time complexity to O (log exponent), making the function fast and suitable for applications in cryptography, number theory, and modular arithmetic.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding Exponentiation by Squaring: Many students struggle to understand why squaring the base and halving the exponent gives the same result as normal repeated multiplication. The binary idea behind it is not immediately intuitive.

2. Handling Odd vs Even Exponents: When to multiply the result with the base. When to only square the base. This leads to incorrect outputs.

3. Forgetting to Apply Modulo Frequently Some forget to use: % modulus at each step. Without frequent modulo, numbers become extremely large and slow, or cause overflow in other languages.

4. Misunderstanding the Purpose of Modulo: Students sometimes think modulo is applied only at the end, but efficient modular exponentiation requires applying modulo during every operation to keep values small.

5. Difficulty Debugging Large Exponents: When exponents are very large, it becomes hard to manually verify and debug the steps.

**SKILLS ACHIEVED**:

1. Logical Thinking & Algorithmic Reasoning: Students learn to break down a big mathematical operation into smaller, efficient steps using binary logic and pattern recognition.

2. Understanding of Modular Arithmetic how modulo works, why modulo is applied repeatedly, how it keeps numbers manageable and efficient. This is essential for number theory and cryptography.

3. Mastery of Exponentiation by Squaring: RSA encryption, Fast computations, Competitive programming

4. Efficient Coding Practices: reducing time complexity from O(n) to O (log n), writing optimized loops avoiding overflow and unnecessary operations.

5. Debugging and Analytical Skills: trace algorithm steps, check odd/even conditions, handle large inputs, prevent logical errors in loops.

6. Strong Mathematical–Programming Connection: algorithms, competitive coding, cryptography projects, problem-solving exams.

**Practical No: 21**

**Date: 16-11-25**

**TITLE**: To find Modular Multiplicative Inverse.

**AIM/OBJECTIVE(s)**: **Write a function Modular Multiplicative Inverse mod_inverse (a, m) that finds the number x such that (a * x) ≡ 1 mod m.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

To find the modular multiplicative inverse of a number a modulo m, we use the Extended Euclidean Algorithm (EEA) because: a ^ {-1} \mod m \text {exists only if} \ gcd (a, m) = 1. The EEA helps solve the equation: a x + m y = \gcd (a, m), If gcd = 1, then a x + m y = 1. The value of x obtained from EEA is the modular inverse, and we take: x \mod m

Steps in the Methodology:

1. Check coprimality: Ensure; otherwise, inverse does not exist.

2. Apply Extended Euclidean Algorithm: Recursively compute gcd along with coefficients (x, y) such that: a x + m y = 1

3. Extract modular inverse: The value of x obtained from EEA is the inverse. Adjust by modulus to ensure it is positive: x = x \mod m

4. Return the inverse.

**TOOLS:**

1. Extended Euclidean Algorithm (EEA): A mathematical tool to compute gcd (a, b) and the coefficients x, y satisfying ax + by = \gcd (a, b)

2. Python Programming Language: Recursive functions, Modulo arithmetic, Mathematical operations

3. Mathematical Theorems Used, Bezout's Identity: If, then there exist integers x and y such that ax + my = 1. Modular Arithmetic Rules Ensuring the final answer is within the range to.

**BRIEF DESCRIPTION**:

The mod_inverse (a, m) function calculates the number x such that: (a times x) equiv 1 mod m. To find this value, the function uses the Extended Euclidean Algorithm, which determines integers x and y that satisfy: a x + m y = gcd (a, m) If the gcd (a, m) = 1, the value of x obtained is the modular inverse of an under-modulus m. The function returns x mod m to ensure the result is positive and within the valid range. If the gcd is not 1, then the inverse does not exist and the function returns None.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- Understanding why gcd (a, m) must be 1.
- Confusion between "inverse under multiplication" vs normal division.
- Implementing Extended Euclidean Algorithm recursion.
- Forgetting to take x mod m to make it positive.

**SKILLS ACHIEVED**:

- Number theory fundamentals
- Modular arithmetic
- Recursive algorithm design
- Understanding Extended Euclidean Algorithm
- Mathematical reasoning behind inverse mod

**Practical No: 22**

**Date: 16-11-25**

**TITLE**: To verify the Chinese remainder theorem.

**AIM/OBJECTIVE(s)**: **Write a function Chinese Remainder Theorem Solver crt (remainders, moduli) that solves a system of congruences x ≡ ri mod mi.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Compute the Product of All Moduli
   Calculate $M = m_1 \times m_2 \times \ldots \times m_n$, the product of all moduli.
2. Calculate Each Partial Product
   For each $i$, let $M_i = M/m_i$.
   This means $M_i$ is the product of all moduli except $m_i$.
3. Find Modular Inverses
   For each $i$, compute the modular inverse of $M_i$ modulo $m_i$. The modular inverse $y_i$ satisfies
   $M_i \cdot y_i \equiv 1 (\mathrm{mod} m_i)$.
   Typically, the extended Euclidean algorithm is used to efficiently find these inverses.
4. Combine Results
   Compute $x = \sum_{i=1}^{n} \quad r_i \cdot M_i \cdot y_i$.
   The final solution is given by $x (\mathrm{mod} M)$, ensuring that all original congruence conditions hold.

**Tools:**

Modular Arithmetic: All calculations (such as finding remainders and inverses) are performed using modular arithmetic to ensure correctness with respect to each modulus in the system. Extended Euclidean Algorithm. The core tool for computing modular inverses is the extended Euclidean algorithm. Given two integers $a$ and $b$, the algorithm efficiently

finds integers $x$ and $y$ such that $ax + by = \gcd(a, b)$. When $a$ and $b$ are coprime, $x$ (modulo $b$) is used as the modular inverse of $a$ mod $b$.

List and Product Operations

- Pythons reduce and zip functions help perform product calculations ($M$) and pair remainders with moduli during the summation process.
- Product calculation is key for finding $M = m_1 \times m_2 \times \ldots \times m_n$ as well as each partial product $M_i = M/m_i$.

**BRIEF DESCRIPTION**:

This code efficiently solves a system of simultaneous modular congruences using the Chinese Remainder Theorem. It takes as input two lists—one containing remainder, and one containing modulus that are pairwise coprime—and computes the unique solution $x$ modulo the product of the moduli such that $x \equiv r_i (\mathrm{mod} m_i)$ for each given pair.

**Key Features:**

- Calculates the overall product of all moduli ($M$).
- For each congruence, determines a suitable multiplier and its modular inverse to combine individual solutions.
- Uses the extended Euclidean algorithm to find modular inverses efficiently.
- Aggregates the results to output the smallest solution $x$ satisfying all the given modular equations and the modulus $M$.

This makes the code a practical tool for finding integer solutions to systems of congruences in a mathematically sound and efficient manner.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- **Understanding Modular Inverses**
  Many students find it confusing to compute modular inverses, especially when using the extended Euclidean algorithm, as it requires a good grasp of number theory and modular arithmetic.
- **Pairwise Coprimality Requirement**
  Forgetting to verify that all moduli are pairwise coprime can result in incorrect solutions or unsolvable systems. If this requirement isn't checked, the code can produce meaningless results for some input.
- **Handling Large Numbers**
  As the product of all moduli ($M$) may become very large, arithmetic operations can be computationally intensive, and managing integer overflow or precision issues is challenging in languages that don't handle big integers automatically.
- **Algorithm Implementation Errors**
  Implementing the extended Euclidean algorithm or the summing formula for the CRT can be error-prone due to index mistakes, incorrect variable initialization, and confusion about which values to use for multiplication or modular reduction.
- **Validating Inputs and Outputs**
  Students often struggle with testing their code for all edge cases, such as negative remainders or moduli, verifying the uniqueness of the solution, and wrapping the result correctly by taking $x \bmod M$.

**SKILLS ACHIEVED**:

- **Understanding of Modular Arithmetic**
  Students gain practical insight into modular calculations, which are fundamental in fields like cryptography, coding theory, and algorithms.
- **Algorithmic Problem Solving**
  The process involves breaking down a complex problem into subproblems, enhancing divide-and-conquer and systematic thinking skills. This methodology is widely applicable in computer science.
- **Use of Extended Euclidean Algorithm**
  By implementing modular inverses, students encounter number theory concepts and reinforce their understanding of the greatest common divisor and its computation.
- **Efficient Coding Practices**
  Handling large numbers and managing pairwise coprimality checks helps build robust coding habits, including the use of language-specific libraries for high-precision computations.
- **Debugging and Testing**
  Students improve their ability to debug mathematical algorithms, manage edge cases, and write thorough test cases for their implementations

**Practical No: 23**

**Date: 16-11-25**

**TITLE**: To verify quadratic residue.

**AIM/OBJECTIVE(s)**: **Write a function Quadratic Residue Check is_quadratic_residue (a, p) that checks if x2 ≡ a mod p has a solution.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

- Euler's Criterion states: For an odd prime $p$ and integer $a$ with $\gcd(a,p) = 1$,
  $a$ is a quadratic residue modulo $p$ if and only if

$$a^{\frac{p-1}{2}} \equiv 1 (\mathrm{mod} p)$$

Otherwise, it is a quadratic non-residue and

$$a^{\frac{p-1}{2}} \equiv -1 (\mathrm{mod} p)$$

- The code applies this by computing $a^{(p-1)/2} \mathrm{mod}\ p$ using efficient modular exponentiation (pow function in Python).
- If the result is $1$, the function returns True indicating a solution $x$ exists for $x^2 \equiv a (\mathrm{mod} p)$; if it is any other value (specifically $p-1$ which is congruent to $-1$), it returns False.
- This method leverages Fermat's Little Theorem and properties of primes to quickly determine quadratic residue status without explicitly finding $x$.

**TOOLS:**

The main tool used in this code is Python's built-in pow () function with three arguments: pow (base, exponent, modulus). This function efficiently performs modular exponentiation, calculating

$$a^{(p-1)/2} \mathrm{mod}\ p$$

in logarithmic time complexity using an optimized algorithm.

**BRIEF DESCRIPTION**:

This code determines if a given number $a$ is a quadratic residue modulo a prime $p$, meaning it checks if the congruence $x^2 \equiv a(\mathrm{mod}p)$ has a solution. It uses Euler's criterion, which reduces the problem to computing $a^{(p-1)/2}\mathrm{mod}\ p$ efficiently via modular exponentiation. If the result is 1, the number is a quadratic residue; otherwise, it is not. This approach quickly verifies the existence of square roots modulo $p$ without explicitly finding them, making it efficient and mathematically grounded in number theory.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding Euler's Criterion: Grasping the theorem and why $a^{(p-1)/2} \equiv 1(\mathrm{mod}p)$ implies $a$ is a quadratic residue requires solid background in number theory.
2. Modular Exponentiation Concept: Comprehending how modular exponentiation works and why it efficiently replaces direct computation of powers modulo a prime can be challenging.
3. Correct Usage of Python's pow () Function: Knowing that pow (a, b, c) does modular exponentiation and using it properly is sometimes confusing for beginners.
4. Handling Edge Cases: Recognizing that if $a \equiv 0(\mathrm{mod}p)$, it is trivially a residue, and managing inputs outside valid ranges or primes requires careful input validation.
5. Debugging Mathematical Code: Differentiating between coding errors and misunderstandings of mathematical properties can complicate troubleshooting.

6. Prime Modulus Requirement: Understanding that the method works reliably only when $p$ is prime, and why non-prime modulus cases are different, can be subtle.

**SKILLS ACHIEVED**:

1. Number Theory Understanding: Gain grasp of foundational concepts like quadratic residues, modular arithmetic, and Euler's criterion.
2. Modular Exponentiation: Learn how to efficiently compute powers modulo a prime using built-in functions like Python's pow ().
3. Algorithmic Thinking: Develop problem-solving approaches to translate mathematical theorems into computational algorithms.
4. Code Optimization: Understand how applying mathematical properties reduces computational complexity versus brute force.
5. Debugging Mathematical Code: Identify and fix errors at the intersection of abstract math and practical programming.
6. Mathematical Rigor: Appreciate rigorous proofs' role in confirming algorithm correctness and reliability.
7. Preparation for Advanced Topics: Build a foundation useful for more advanced studies in cryptography, coding theory, and computational number theory

**Practical No: 24**

**Date: 16-11-25**

**TITLE**: To find order mod.

**AIM/OBJECTIVE(s)**: **Write a function order mod (a, n) that finds the smallest positive integer k such that ak ≡ 1 mod n.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Check Coprimality: First, ensure that $a$ and $n$ are coprime (gcd $(a,n) = 1$). Without this, the order is not defined.
2. Iterative Modular Multiplication: Start with $k = 1$ and calculate $a^k \bmod n$.
3. Repeat Until 1: Increment $k$ and keep computing $a^k \bmod n$ until the result is 1.
4. Return the Smallest $k$: The first $k$ for which $a^k \equiv 1 (\bmod n)$ is the multiplicative order.

**TOOLS:**

- math.gcd (): Efficiently determines if $a$ and $n$ are coprime before proceeding to find the order.
- Modular arithmetic: The code performs modular multiplication iteratively using Python's modulo operator %.
- Basic control flow: Loops and conditionals systematically compute powers until the congruence $a^k \equiv 1 (\bmod n)$ is met.

**BRIEF DESCRIPTION**:

This code finds the smallest positive integer $k$ such that $a^k \equiv 1 (\bmod n)$ (the order of $a$ modulo $n$). It first checks if $a$ and $n$ are coprime using the greatest common divisor (gcd) function. Then, it iteratively computes powers of $a$ modulo $n$ by multiplying the current value by $a$ and taking the modulus until the result is 1. The number of

iterations $k$ taken to reach 1 is returned as the multiplicative order. This brute-force approach systematically tests increasing exponents until it finds the smallest one satisfying the congruence. The code leverages Python's modulo operator % to keep the computation within bounds, ensuring efficient calculation of modular powers. This approach directly applies modular arithmetic concepts and guarantees finding the minimal $k$ for which the modular exponentiation yields 1.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding Modular Arithmetic: Grasping the concept of numbers "wrapping around" and why modular exponentiation works can be non-intuitive.
2. Handling Negative Numbers: Ensuring correct handling of negative values in modulo operations to keep results in the expected range is tricky for many beginners.
3. Efficient Computation: Knowing how to efficiently compute powers modulo $n$ to avoid large intermediate results and potential overflow may be challenging.
4. Algorithm Implementation: Translating the mathematical definition of multiplicative order into iterative or optimized algorithms requires good programming and mathematical insight.
5. Debugging Subtle Errors: Modular arithmetic operations can sometimes yield unexpected results due to operator precedence or misunderstandings about modular properties, making debugging harder.

6. Edge Cases and Input Validation: Failing to validate inputs such as ensuring $a$ and $n$ are coprime or handling small or large values correctly

**SKILLS ACHIEVED**:

1. Understanding and applying modular arithmetic concepts
2. Mastering the use of Python's gcd function for coprimality checks
3. Implementing iterative algorithms for modular exponentiation
4. Developing problem-solving skills in computational number theory
5. Gaining experience with control flow structures like loops and conditionals in programming
6. Learning to handle edge cases and validate mathematical conditions programmatically
7. Enhancing debugging abilities in mathematical code contexts
8. Building foundations for more advanced topics in cryptography and algorithms involving modular groups

**Practical No: 25**

**Date: 16-11-25**

**TITLE**: To find Modular Exponentiation number.

**AIM/OBJECTIVE: Write a function Fibonacci Prime Check is_fibonacci_prime(n) that checks if a number is both Fibonacci and prime.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Fibonacci Check Using a Mathematical Property:
   Instead of generating Fibonacci numbers, the code uses the property that a number $n$ is Fibonacci if either $5n^2 + 4$ or $5n^2 - 4$ is a perfect square. This offers a constant-time check.
2. Prime Check Using Efficient Primality Tests:
   The code performs a primality test, such as trial division up to $\sqrt{n}$, to determine if the number is prime.
3. Combined Verification:
   The number is classified as a Fibonacci prime only if it satisfies both the Fibonacci property and the primality condition.
4. Optimization Options:
   For larger numbers, sieve algorithms or probabilistic primality tests (e.g., Miller-Rabin) may be used alongside efficient Fibonacci membership checks.

**TOOLS:**

1. Mathematical property checks for Fibonacci numbers using the perfect square test of $5n^2 + 4$ or $5n^2 - 4$.
2. Primality testing algorithms like trial division, sieve of Eratosthenes, or probabilistic tests (e.g., Miller-Rabin) to efficiently check if a number is prime.

3. Prime sieves such as the Sieve of Eratosthenes to generate all primes up to a limit quickly before testing Fibonacci conditions.

**BRIEF DESCRIPTION**:

This code checks if a given number $n$ is both a Fibonacci number and a prime number. It does so by:

1. Checking if $n$ is a Fibonacci number using the mathematical property that $n$ is Fibonacci if and only if one or both of $5n^2 + 4$ or $5n^2 - 4$ is a perfect square.
2. Checking if $n$ is prime through an efficient primality test (e.g., trial division up to $\sqrt{n}$).
3. Returning True only if both conditions hold, indicating $n$ is a Fibonacci prime; otherwise, it returns False.

The code leverages number theory properties for fast verification without generating the entire Fibonacci sequence. It combines mathematical insight and algorithmic checks into a concise program to identify Fibonacci primes effectively

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Trouble with perfect square checking: Use floating-point square root and get rounding errors. Forget to convert sqrt result to integer before squaring. Don't understand why perfect-square checking works

2. Prime checking mistakes: Checking divisibility up to n instead of √n (very slow). Missing edge cases like n = 0, 1, or negative numbers. Not handling even numbers properly.

3. Mixing up the order of checks: Some students check Fibonacci first and then prime, leading to unnecessary calculations. Prime checking should come first because it eliminates most numbers quickly.

4. Logical mistakes: Using OR instead of AND (prime OR Fibonacci). Returning the wrong Boolean values. Wrong indentation or misplaced return statements

5. Not understanding the concept of Fibonacci prime: Fibonacci number, Prime number, Fibonacci prime (a number that is both)

**SKILLS ACHIEVED**:

1. Understanding of Number Theory: Prime numbers, Fibonacci numbers, Perfect-square properties, Mathematical pattern recognition

2. Efficient Algorithm Design: Use the √n method for prime checking. Apply formulas instead of brute-force loops. Avoid unnecessary calculations. Write optimized and clean code.

3. Logical Thinking and Condition Handling: Break a big problem into smaller functions. Combine logical conditions using AND/OR. Handle edge cases correctly (0, 1, negatives). Structure code with clarity and purpose

4. Modular Programming Skills: Create helper functions (is_prime, is_fibonacci). Use modular code for reusability. Understand the benefits of separating logic

5. Mathematical Problem-Solving Skills: Convert math formulas into code. Use the perfect-square test to detect Fibonacci numbers. Apply mathematical reasoning to programming tasks

6. Debugging and Error Handling: Identifying logical errors. Testing edge cases. Understanding common mistakes. Ensuring the function gives accurate results

**Practical No: 26**

**Date: 16-11-25**

**TITLE**: To generate Lucas number.

**AIM/OBJECTIVE(s)**: **Write a function Lucas Numbers Generator lucas_sequence(n) that generates the first n Lucas numbers (similar to Fibonacci but starts with 2,1).**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Handle small values of n: If n ≤ 0, return an empty list, If n = 1, return [2], If n = 2, return [2, 1]. This ensures the function works correctly for edge cases.

2. Initialize the starting list: Lucas = [2, 1], This forms the base for generating further terms.

3. Use a loop to generate remaining terms for index i from 2 to n−1: Compute the next Lucas number L_i = L_{i-1} + L_{i-2} Append it to the list. This uses the recurrence relation efficiently.

4. Return the final list: After the loop completes, return the list containing the first n Lucas numbers.

**TOOLS:**

A Python list is used to store and build the Lucas sequence: Lucas = [2, 1]

Lists allow: Dynamic appending

Index-based access (lucas[i-1], lucas[i-2])

3. Looping (for loop): A for loop is used to generate each Lucas number after the first two

for i in range (2, n):

next_value = lucas[i-1] + lucas[i-2], Loops help automate repetitive addition.

**BRIEF DESCRIPTION**:

The function lucas_sequence(n) generates the first n Lucas numbers, which form a mathematical sequence similar to the Fibonacci series but starting with 2 and 1 instead of 0 and 1. The code first handles small input cases like n = 0, 1, or 2. Then it initializes a list with the first two Lucas numbers: [2, 1]. After that, it uses a loop to calculate each new Lucas number by adding the previous two numbers:

$L_n = L_{n-1} + L_{n-2}$

Each newly computed term is appended to the list.

Finally, the function returns the entire list of Lucas numbers.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Confusion between Fibonacci and Lucas sequences: Fibonacci starts with 0, 1. Lucas starts with 2, 1. They may accidentally use the wrong starting values.

2. Forgetting to handle small values of n: Not checking for n = 0 or negative values. Returning wrong outputs for n = 1 or n = 2

3. Errors in using list indices: Use wrong index positions. Forget that list indexing starts from 0. Cause index-out-of-range errors

Example mistake:

```
lucas[i] = lucas[i-1] + lucas[i-2]   # wrong
```

4. Wrong loop range: for i in range(n):  # wrong

instead of starting at 2: for i in range (2, n):  # correct

5. Misunderstanding recursion vs iteration: It becomes complicated, slow for large n, causes stack overflow, iteration is simpler and faster.

6. Returning the wrong value: Only the last Lucas number, extra values, an empty list for valid inputs, correct output must be a list of first n Lucas numbers.

**SKILLS ACHIEVED**:

1. Understanding Mathematical Sequences: what Lucas numbers are, how they relate to Fibonacci numbers, how recurrence relations work, how sequences grow and are generated step-by-step

2. Algorithmic Thinking: Break the problem into smaller parts, identify base cases (n = 0, 1, 2), use loops to generate future terms, apply recurrence formulas in code

3. Handling Edge Cases: Negative inputs n = 0, n = 1 or n = 2. This improves defensive programming skills.

4. Problem-Solving and Debugging: Identifying logical errors, fixing index mistakes, understanding correct loop ranges, testing with sample inputs

5. Modular Code Design by separating initialization from looping, students understand: Organizing code cleanly, making functions easier to read and maintain

**Practical No: 27**

**Date: 16-11-25**

**TITLE**: To find perfect power number.

**AIM/OBJECTIVE(s)**: **Write a function for Perfect Powers Check is_perfect_power(n) that checks if a number can be expressed as ab where a > 0 and b > 1.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. **.** Understand the range of $b$: The exponent $b$ only needs to be checked from 2 up to $\log_2(n)$. This is because if $b$ is larger than $\log_2(n)$, then $a^b$ would exceed $n$ for any $a \geq 2$.
2. For each candidate $b$, determine if $n$ is a perfect $b$-th power:
    - Compute the $b$-th root of $n$, which can be approximated using Newton's method or binary search.
    - Round this root to the nearest integer $a$.
    - Check if $a^b = n$.
    - If yes, $n$ is a perfect power and return True.
    - If no for all $b$, return False.
3. To find the $b$-th root of $n$ efficiently:
    - Use numerical methods such as Newton's method for root approximation.
    - Alternatively, use binary search to find the integer root.
4. Limitations:
    - For large numbers, precise calculation of roots may require careful handling.
    - Only integer values for $a$ and $b$ are considered

**TOOLS:**

1. This process relies on numerical root approximation and integer power verification.
2. The loop upper bound is logarithmic to reduce the number of checks.
3. This approach has efficient implementations in many programming languages and can handle large $n$.

**BRIEF DESCRIPTION**:

The function is_perfect_power(n) checks whether a given number n can be written in the form $a^b$, where a > 1 and b > 1. It works by testing all possible exponents from 2 up to $\log_2(n)$, computing the corresponding base using the b-th root of n, and verifying if raising that base to the exponent reproduces n exactly. If any such pair (a, b) satisfies the condition, the number is identified as a perfect power; otherwise, it is not.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Misunderstanding the definition of perfect power.
2. Forgetting to limit the exponent using $\log_2(n)$.
3. Floating-point precision issues when taking roots.
4. Not considering rounding when computing roots.
5. Including 1 or 0 incorrectly as perfect powers.

**SKILLS ACHIEVED**:

1. Understanding logarithms and roots in computation.
2. Handling floating-point errors effectively.
3. Working with mathematical number-theory functions.
4. Looping strategically to reduce unnecessary computation.
5. Implementing power checks using Python efficiently.

**Practical No: 28**

**Date: 16-11-25**

**TITLE**: To verify Collatz sequence length.

**AIM/OBJECTIVE(s)**: **Write a function Collatz Sequence Length collatz_length(n) that returns the number of steps for n to reach 1 in the Collatz conjecture.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Use a loop that repeatedly applies the transformation to $n$.
2. Keep track of the count of steps.
3. End when $n = 1$.
4. Return the step count as the length of the Collatz sequence for $n$.

**TOOLS:**

1. Basic Python arithmetic
2. Loops (while loop)
3. Conditionals (if-else statements)

**BRIEF DESCRIPTION**:

The provided Collatz sequence length code follows the rules of the Collatz conjecture: for a positive integer $n$, if it is even, the code divides it by 2; if odd, it multiplies by 3 and adds 1. This process repeats until $n$ becomes 1. The code keeps a count of how many such steps are performed, which represents the sequence length or number of steps to reach 1. The overall goal is to determine how long it takes for $n$ to transition through the sequence defined by these operations down to the terminating value 1. It's a straightforward iterative implementation of the Collatz process that applies the simple conditional transformation repeatedly while tracking the step count, answering the conjecture's fundamental question: how many steps does it take to reach 1 for a given starting number? This serves as both a computational demonstration and a tool for analysis of Collatz sequences.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- Forgetting to increment the step counter.
- Infinite loops if they do not correctly update n.
- Using / instead of //, causing floats instead of integers.
- Not handling input validation (like negative numbers).

**SKILLS ACHIEVED**:

- Understanding of loops and conditions.
- Logical sequence design.
- Mathematical reasoning with number theory.
- Writing clean and efficient functions.

**Practical No: 29**

**Date: 16-11-25**

**TITLE**: To find polygonal number.

**AIM/OBJECTIVE(s)**: **Write a function Polygonal Numbers polygonal_number (s, n) that returns the n-th s-gonal number.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Understand the Concept: Polygonal numbers generalize triangular, square, pentagonal, hexagonal numbers, etc. Each polygonal number represents dots arranged in the shape of an s-sided polygon. Examples:

- s = 3 → Triangular numbers
- s = 4 → Square numbers
- s = 5 → Pentagonal numbers

2. Use the General Formula the formula for the n-th s-gonal number is: P(s, n) = \frac{(s - 2)n^2 - (s - 4)n}{2}. This formula works for all polygonal series.

3. Implement in Python using integer arithmetic: def polygonal_number(s, n), return ((s - 2) * n * n - (s - 4) * n) // 2

4. Validate with Known Cases triangular number (s = 3):

n = 5 → 15 ✓

Square number (s = 4):

n = 4 → 16 ✓

Pentagonal number (s = 5):

n = 3 → 12 ✓

5. Ensure Function is General: The function must support any s ≥ 3 and n ≥ 1, making it reusable for all polygonal sequences.

**TOOLS:**

1. Arithmetic Operators Essential Python operators used:

- * for multiplication
- - for subtraction
- for integer division

These directly translate the mathematical formula into code.

2. Function Definition

Using Python's def keyword to create reusable functions: def polygonal_number(s, n).

**BRIEF DESCRIPTION**:

The polygonal_number(s, n) function calculates the n-th s-gonal number, which represents a number that can be arranged in the shape of a polygon with s sides (triangle, square, pentagon, etc.). It uses the general mathematical formula: P(s, n) = \{(s - 2)n^2 - (s - 4)n}{2}. This formula works for all polygonal sequences, including triangular, square, pentagonal, hexagonal, and beyond. By simply changing the value of s, the function can generate any type of polygonal number. The function is efficient, uses basic arithmetic operations, and returns an integer result. It is useful in number theory, sequences, and mathematical pattern analysis.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding the General Formula Students may struggle to understand how one formula works for all polygonal numbers. Terms like and may seem abstract without visual diagrams.

2. Differentiating Between Types of Polygonal Numbers Students sometimes confuse:

- Triangular numbers (s = 3)
- Square numbers (s = 4)
- Pentagonal numbers (s = 5)
- Hexagonal numbers (s = 6)

Remembering which value of s corresponds to which shape can be confusing at first.

3. Translating Math Formula → Code

Even when students understand the formula, they may struggle with, properly using parentheses. Using integer division (// 2) instead of float division.

4. Logical Errors in Implementation Common mistakes include:

- Forgetting parentheses → wrong order of operations
- Using / instead of // → causes floating-point results
- Misplacing (s - 4) term
- Not validating inputs (s < 3)

5. Difficulty Visualizing Polygonal Patterns Students often find it hard to visualize how dots form shapes like pentagons or hexagons, making the formula feel less intuitive.

6. Confusing Polygonal Numbers with Figurate Numbers Polygonal numbers are a subset of figurate numbers.

**SKILLS ACHIEVED**:

1. Translating Math to Code: They gain experience converting a mathematical equation into executable python code, using arithmetic operators, managing order of operations, handling integer division

2. Working With Sequences: Students learn how number sequences are generated and how formulas relate to patterns in number theory.

3. Function Design in Python By implementing the function, students practice: Defining functions with parameters, Returning computed results, Writing clean, reusable code.

4. Problem solving skills: Break a complex formula into smaller steps. Test the function using known values. Troubleshoot incorrect outputs

5. Computational Thinking Students develop skills in:

- Algorithmic thinking
- Applying formulas efficiently
- Understanding generalization (one formula → many shapes)

6. Improved Confidence in Number Theory Working with polygonal numbers deepens understanding of: Patterns, Figurate numbers, Mathematical structure and symmetry

**Practical No: 30**

**Date: 16-11-25**

**TITLE**: To verify Carmichael number.

**AIM/OBJECTIVE(s)**: **Write a function Carmichael Number Check is_carmichael(n) that checks if a composite number n satisfies an−1 ≡ 1 mod n for all a coprime to n.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Verify $n$ is composite: Carmichael numbers are by definition composite, so if $n$ is prime, it cannot be Carmichael.

2. For each integer $a$ coprime to $n$: Compute $a^{n-1} \bmod n$ using modular exponentiation methods. Check if $a^{n-1} \equiv 1 (\bmod n)$. If for any coprime $a$, this condition fails, $n$ is not Carmichael.
3. To optimize, instead of checking all $a$, tests can use: Korselt's criterion which states $n$ is Carmichael if and only if. $n$ is square-free (no repeated prime factors). For every prime divisor $p$ of $n$, $p - 1$ divides $n - 1$. First factorize $n$ and verify these conditions to confirm if it is Carmichael.
4. Algorithms may use a combination of Fermat tests for multiple bases $a$ and prime factorization checks (Korselt's) for computational efficiency.
5. Use efficient modular exponentiation (e.g., fast exponentiation by squaring) to perform $a^{n-1} \bmod n$.
6. Halt and return False immediately if any $a$ does not satisfy the condition, else return True after exhaustively checking all coprime $a$ or verifying Korselt's criterion.

**TOOLS:**

- Korselt's Criterion : Main method to detect Carmichael numbers
- Miller–Rabin: Check if number is prime
- Pollard's Rho: Factorize the number
- GCD (Euclid Algorithm): Needed for Pollard Rho & coprimality
- Modular Arithmetic: Backbone of primality tests
- Python libraries (math, random, Counter): Implementation helpers

**BRIEF DESCRIPTION**:

The function is_carmichael(n) checks whether a composite number is a Carmichael number, which is a special type of number that behaves like a prime in Fermat's little theorem. Instead of testing the condition a^{n-1} \equiv 1 \mod {n}. Using this criterion, the function first checks that is composite, then factorizes using Pollard's Rho algorithm. If every prime factor of appears only once (i.e., is square-free) and satisfies: (p - 1) \mid (n - 1). This method is much faster than brute-force checking and allows the function to detect Carmichael numbers efficiently, even for moderately large integers.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding What Carmichael Numbers Are Students often struggle with: The idea that Carmichael numbers are composite numbers that behave like primes. Why holds for all coprime? This abstract concept can be confusing without strong number theory background.

2. Complexity of Korselt's Criterion: Although Korselt's criterion simplifies the test, students may find it difficult to apply Understanding square-free condition Checking (p − 1) divides (n − 1) Connecting prime factor properties to Carmichael behaviour.

3. Factoring Large Numbers: To apply the criterion, students must factor the number. Common challenges:

- Basic trial division is slow
- Pollard's Rho algorithm can feel advanced or confusing
- Difficulty understanding randomness and iteration in factorization

4.Implementing Primality Tests Students may face problems with: Understanding Miller–Rabin primality test. Writing modular exponentiation efficiently. Avoiding errors in edge cases (e.g., small numbers)

5. Handling Composite vs. Prime Numbers Many students mistakenly: Check Carmichael condition for prime numbers. Forget that Carmichael numbers must be composite

**SKILLS ACHIEVED**:

1. Strong Understanding of Number Theory Students gain knowledge in:

- Composite vs. prime numbers
- Modular arithmetic
- Fermat's Little Theorem
- Special composite numbers (Carmichael numbers)

2. Ability to Apply Korselt's Criterion Students learn how to use a powerful theoretical tool: Checking square-free condition Verifying divisor relations: (p - 1) \mid (n - 1)

3. Practical Experience With Primality Testing Students become comfortable with: Miller–Rabin primality test, Using modular exponentiation for primality checks, Handling probabilistic vs. deterministic tests

4. Skills in Integer Factorization Students learn how to: Factor numbers efficiently, Use advanced algorithms like Pollard's Rho, Understand the role of randomness in factorization

5. Mastery of Modular Arithmetic in Code They gain practice with: Computing efficiently, Using Python's pow(a, b, m), Avoiding overflow and time complexity issues

**Practical No: 31**

**Date: 16-11-25**

**TITLE**: To implement the probabilistic miller-robin test.

**AIM/OBJECTIVE(s)**: **Implement the probabilistic Miller-Rabin test is_prime_miller_rabin (n, k) with k rounds.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

- Check small cases (n < 2, n = 2/3, even numbers).
- Rewrite by factoring out powers of 2.
- Repeat k rounds using random base.
- Compute as the first primality check.
- Square x repeatedly to check if it becomes.
- Fail early if no square reaches → composite.
- If all rounds pass → n is probably prime.

**TOOLS:**

- Modular exponentiation (pow (a, d, n)) for fast computation of.
- Random number generation to pick bases for testing.
- Decomposition of into the form.
- Repeated squaring technique for checking strong pseudoprime conditions.
- Basic arithmetic operations (multiplication, modulo, division by 2).
- Conditional logic to detect composite numbers early based on witness behaviour.

**BRIEF DESCRIPTION**:

The Miller–Rabin primality test is a probabilistic algorithm used to determine whether a number is prime. The method rewrites in the form and then tests the number using randomly chosen bases. For each base, it checks whether the number behaves like a prime through modular exponentiation and repeated squaring. If any base proves the number

composite, the algorithm stops immediately. If all k rounds pass, the number is declared “probably prime”, with the probability of error decreasing as more rounds are performed.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- Understanding the logic behind rewriting.
- Confusion with modular exponentiation and how pow (a, d, n) works.
- Difficulty interpreting why some bases detect compositeness and others don’t.
- Trouble implementing repeated squaring correctly.
- Managing randomness and choosing appropriate values of.
- Misunderstanding the meaning of “probably prime” vs. “definitely composite.”.

**SKILLS ACHIEVED**:

- Applying probabilistic algorithms for primality testing.
- Using modular exponentiation efficiently in code.
- Understanding and implementing the decomposition of.
- Working with repeated squaring in modular arithmetic.
- Gaining experience with randomness in algorithm design.
- Strengthening logical reasoning for composite vs. probable-prime detection.

**Practical No: 32**

**Date: 16-11-25**

**TITLE**: To implement pollard rho number.

**AIM/OBJECTIVE(s)**: **Implement pollard_rho(n) for integer factorization using Pollard's rho algorithm.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Choose a polynomial function (commonly): $f(x) = (x^2 + 1) \% n$
2. Pick a random starting value x = 2, y = 2 (tortoise–hare approach).
3. Repeat:
   - Move x = f(x) (tortoise: one step)
   - Move y = f(f(y)) (hare: two steps)
   - Compute d = gcd(|x – y|, n)
4. If d = 1, continue searching.
5. If 1 < d < n, you found a non-trivial factor.
6. If d = n, restart with a new function/seed.

**TOOLS:**

- Modular arithmetic to compute the polynomial function efficiently.
- GCD (Greatest Common Divisor) to detect shared non-trivial factors.
- Floyd's cycle-finding (Tortoise–Hare) to detect repeating values.

**BRIEF DESCRIPTION**:

Pollard's Rho is a fast probabilistic factorization algorithm that uses a pseudo-random polynomial function, modular arithmetic, and cycle detection to uncover a non-trivial factor of a composite number, making it efficient for large integers with small factors.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

- Understanding why the “tortoise & hare” method finds cycles.
- Handling the case when no factor is found.
- Ensuring the GCD computation is correct.
- Understanding the probabilistic nature (results vary with seeds).

**SKILLS ACHIEVED**:

- You learn probabilistic factorization.
- How to combine GCD, modular arithmetic, and cycle detection.
- Apply number theory to real algorithms used in cryptography.

**Practical No: 33**

**Date: 16-11-25**

**TITLE**: To zeta approx. term.

**AIM/OBJECTIVE(s)**: **Write a function zeta_approx (s, terms) that approximates the Riemann zeta function ζ(s) using the first 'terms' of the series.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

1. Input the complex number $s$ and the number of terms terms.
2. Initialize the sum to zero.
3. Loop over $n$ from 1 to terms:
    - Compute $\frac{1}{n^s}$ (using complex exponentiation for complex $s$)
    - Add the term to the sum.
4. Return the accumulated sum as the approximate value of $\zeta(s)$.

**TOOLS:**

1. Series Summation (Euler-Maclaurin formula): Approximates ζ(s) by summing terms of the infinite series with corrections using Euler-Maclaurin summation to accelerate convergence.
2. Riemann-Siegel Formula: A powerful technique especially for computing ζ(s) on the critical line $\mathrm{Re}(s) = \frac{1}{2}$ for large imaginary parts, reducing the number of terms needed.
3. Gaussian Quadrature and Numerical Integration: Used in some refined approximations involving integrals in error terms.

**BRIEF DESCRIPTION**:

The code approximates the Riemann zeta function $\zeta(s)$ by summing the first 'terms' terms of its defining infinite series:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

The function accepts a complex parameter $s$ and an integer term determining how many terms of the series to sum. It iteratively computes $\frac{1}{n^s}$ for $n = 1$ to terms, accumulating the sum which approximates $\zeta(s)$. This approach works for $\mathrm{Re}(s) > 1$, where the series converges. The approximation improves as more terms are added but is limited by computational time and numerical precision. In essence, the code implements the straightforward partial sum of the Riemann zeta series, demonstrating the function's evaluation using basic complex arithmetic and loops. It serves as a practical introduction to the zeta function's numerical computation before using more advanced methods for efficiency or convergence acceleration.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Complex Arithmetic Understanding: Handling complex numbers and operations like raising a number to a complex power may be unfamiliar or confusing.
2. Convergence Issues: The series converges only for complex numbers with real part greater than 1. Students might struggle to grasp why it fails or diverges outside this region.
3. Computational Efficiency: Summing a large number of terms to get good accuracy can be slow, and students may find it difficult to optimize or decide on a sufficient number of terms.

4. Numerical Precision: Floating-point errors accumulate in complex power calculations and summation, leading to inaccurate results if not carefully managed.
5. Theory vs Implementation Gap: Understanding the theory behind the zeta function — such as analytic continuation or why the series represents the function — can be abstract and challenging to translate into code.
6. Error Handling and Edge Cases: Managing inputs outside the domain of convergence and interpreting outputs realistically requires a deeper understanding that beginners may lack.

**SKILLS ACHIEVED**:

1. Complex Number Arithmetic: Understanding and applying operations on complex numbers, including exponentiation with complex exponents.
2. Looping and Series Computation: Implementing iterative summation of series terms properly.
3. Numerical Approximation: Grasping how infinite series are approximated by finite sums and the associated trade-offs.
4. Use of Mathematical Functions in Code: Translating mathematical expressions directly into programming logic.
5. Error and Precision Awareness: Recognizing convergence limits and floating-point precision issues in numerical calculations.
6. Algorithmic Thinking: Structuring a calculation with input parameters like $s$ and number of terms to control accuracy versus performance.
7. Basic Complex Analysis Insight: Acquiring intuition about functions defined over complex domains and their numerical behaviour.

**Practical No: 34**

**Date: 16-11-25**

**TITLE**: To find partition function.

**AIM/OBJECTIVE(s)**: **Write a function Partition Function p(n) partition_function(n) that calculates the number of distinct ways to write n as a sum of positive integers.**

**METHODOLOGY & TOOL USED**:

**METHODOLOGY:**

The partition function $p(n)$ counts the number of distinct ways to represent the integer $n$ as a sum of positive integers, disregarding order. The methodology to calculate $p(n)$ programmatically often uses dynamic programming based on the recurrence relation:

- $p(0) = 1$ (base case)
- For $k > 0$,

$$p(k) = \sum_{j=1}^{k} p(k-j)$$

adjusted to account for counting partitions without order and duplicates, often implemented with the generating function or using Euler's pentagonal number theorem for an efficient recursive approach.

**TOOLS:**

1. Recurrence Relations: The method implements mathematical recurrences like Euler's pentagonal number theorem or simpler forms that express $p(n)$ in terms of smaller partitions.
2. Memory Structures (Arrays): Arrays or lists store computed partition counts to be reused in iterative computations.
3. Loop Constructs: Nested loops iterate over subproblems, building up the number of partitions step-by-step.

**BRIEF DESCRIPTION**:

The partition function code uses dynamic programming to efficiently calculate the number of distinct ways to express an integer $n$ as a sum of positive integers.

Briefly:

- It initializes an array $dp$ where $dp[i]$ represents the number of partitions of $i$.
- Starts with $dp = 1$ as the base case (there is one way to partition zero).
- Iteratively builds up the solution for all integers from 1 to $n$.
- For each integer $i$, it updates the counts for all sums $j \geq i$ by adding $dp[j - i]$, which means including $i$ as part of the partition.
- By incrementally updating $dp$, it avoids redundant calculations and accumulates the total count of partitions for $n$.
- The final result $dp[n]$ gives the total number of distinct partitions of $n$.

**RESULTS ACHIEVED**:

**DIFFICULTY FACED BY STUDENT**:

1. Understanding the Problem: Grasping that partitioning counts distinct sums without regard to order can be conceptually challenging.
2. Dynamic Programming Concept: Difficulty in understanding how to break the problem into subproblems and use a DP table to store intermediate results.

3. Indexing and Looping Logic: Confusion over the order of loops and how to update the DP array correctly to avoid double-counting partitions.
4. Initialization of Base Cases: Recognizing that $dp = 1$ is essential as the base case representing one way to partition zero.
5. Handling Large Inputs: Managing memory and execution time as $n$ increases, because the DP array can grow large and computations increase.
6. Off-by-One Errors: Mistakes with inclusive/exclusive ranges in loops or indexing the DP array.
7. Debugging Recursive vs Iterative Solutions: Switching from a recursive mathematical understanding to iterative implementation causes conceptual and coding challenges.

**SKILLS ACHIEVED**:

1. Understanding the Problem: Grasping that partitioning counts distinct sums without regard to order can be conceptually challenging.
2. Dynamic Programming Concept: Difficulty in understanding how to break the problem into subproblems and use a DP table to store intermediate results.
3. Indexing and Looping Logic: Confusion over the order of loops and how to update the DP array correctly to avoid double-counting partitions.
4. Initialization of Base Cases: Recognizing that $dp = 1$ is essential as the base case representing one way to partition zero.
5. Handling Large Inputs: Managing memory and execution time as $n$ increases, because the DP array can grow large and computations increase.
6. Off-by-One Errors: Mistakes with inclusive/exclusive ranges in loops or indexing the DP array.
7. Debugging Recursive vs Iterative Solutions: Switching from a recursive mathematical understanding to iterative implementation causes conceptual and coding challenges.